DDCI: Unsupervised Domain Adaptation for Remote Sensing Images Based on Diffusion Causal Distillation

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***Abstract*— The distribution of remote sensing (RS) images can vary significantly due to seasonal changes and lighting conditions, making it difficult for deep learning models to generalize effectively across different RS datasets. This variation leads to a domain gap that hampers model performance when applied to new, unseen data. To tackle this challenge, we introduce DDCI, a novel unsupervised domain adaptation (UDA) framework designed to bridge the domain gap in RS image perception. Our framework consists of two key components, i.e., the adaptation diffusion distillation (ADD) module and the consistent causal intervention (CCI) module. The ADD module addresses the domain gap by aligning the source and target domains. It enhances the representation of the target domain by distilling semantic knowledge from the teacher model of the source domain. This process allows the target domain to benefit from the rich features of the source domain, leading to improved model generalization. The CCI module focuses on removing spurious correlations between domain-agnostic knowledge and domain-specific knowledge. By carefully considering the distinct characteristics of the target domain while preserving the specificity of the source domain, the CCI module ensures that only relevant, causal information is transferred between domains. This prevents overfitting to irrelevant domain-specific features and enhances model robustness. We demonstrate the effectiveness of the DDCI framework on RS scene classification tasks, utilizing four widely recognized RS datasets. Our results show significant performance improvements, underscoring the potential of this approach to boost the adaptability of deep learning models across diverse RS image datasets.**

***Index Terms*— Causal inference, diffusion model, remote sensing (RS), scene classification, unsupervised domain adaptation (UDA).**

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I. Introduction

THE continuous advancement of sensors and remote sensing (RS) platforms has led to an abundance of RS data. Currently, deep learning models become the most popular tools for RS image interpretation [1], [2], [3]. Benefiting from the abundance of RS data, these models have shown remarkable results as they can learn RS image representation and inference in an end-to-end manner. However, their success relies on two impractical assumptions, i.e., a large amount of labeled data is available during training, and the training and testing data follow an independent and identically distributed (i.i.d.). It is time-consuming and laborious to obtain high-quality labeled data, and professional staff in the community of RS are required. The performance degrades when the model is applied for new RS data even when a model is trained with sufficient labeled data. Models are susceptible to variations in data distribution between the source and target domains (i.e., domain gap). Therefore, the performance for target domains of a trained model on source domains can significantly degrade when encountering out-of-distribution (OOD) data.

Unsupervised domain adaptation (UDA) methods for RS images are proposed as a solution to tackle this problem, aiming to bridge the knowledge gap by transferring learning from a labeled source domain to unseen and unlabeled target domains [4]. To lessen the domain disparity throughout the adaptation phase, a multitude of UDA approaches concentrate on acquiring features that remain consistent across domains, either explicitly or implicitly, thereby reducing the divergence in distributions between the source and target domains.

Early research focused on feature alignment by matching interdomain statistics to reduce distribution discrepancies. DDC [5] aimed to maximize domain-invariant features, while DCORAL [6] refined feature distribution through correlation alignment. Later, multirepresentation learning emerged to capture richer cross-domain information: MRAN [7] enhanced domain invariance through multiscale feature fusion, and AMRAN [8] leveraged attention mechanisms for robust feature extraction. For complex RS scenarios, WPS-DSA [9] introduced a weak pseudosupervised decorrelation framework to reduce redundant information. However, these methods often rely on complex alignment strategies, making them sensitive to noise and domain biases.

Adversarial learning employs domain discriminators and feature extractors to achieve implicit alignment.

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RevCGrad [10] pioneered the gradient reversal mechanism for adversarial training. JAN [11] optimized joint multilayer features using joint maximum mean discrepancy (JMMD), while DASNET [12] and TADA [13] focused on task-specific classifiers and transferable attention, respectively. MRDAN [14] further combined dynamic multirepresentation alignment and adversarial training, achieving notable results in RS. However, these methods often face training instability and depend heavily on domain knowledge, leading to potential negative transfer.

Contrastive learning optimizes feature spaces by leveraging positive and negative sample relationships. DAN [15] aligned interdomain conditional distributions using maximum mean discrepancy (MMD). Knowledge distillation (KD) methods, such as SRKT [16], transferred semantic knowledge through adversarial, contrastive, and distillation techniques for RS scene recognition. Despite their effectiveness, these approaches often require complex training strategies and high computational costs. Recently, hybrid methods have emerged, integrating feature alignment, KD, adversarial learning, and contrastive optimization to improve domain adaptation with enhanced accuracy and robustness.

However, due to the need for complex training schemes or task-specific priors, these methods are difficult to apply to various feature types [17]. Directly aligning these mismatched features between the source domain and the target domain can interfere with optimizing the model on the target domain and weaken its performance [18], [19], [20]. Therefore, we hope to learn precious and good features with limited capacity or training methods. We consider feature differences as internal noise in the target domain and hope to generate target-domain features that are very similar to the corresponding source-domain features through the denoising process to ensure that distillation can be performed more consistently. However, this distillation only respects the specificity of the source domain and does not consider the inherent distribution in the target domain. In addition, incorporating domain prior knowledge during training can cause the model to develop biased inductive assumptions. Thus, we hope to eliminate the confusion caused by domain prior knowledge and simultaneously consider the distribution of the source and target domains. To address these problems, we propose a novel UDA framework called DDCI. This framework is composed of two essential components: the adaptation diffusion distillation (ADD) module and the consistent causal intervention (CCI) module.

Inspired by the ability of the diffusion model to transform data distributions, we apply it to the domain adaptation of RS images, where large distribution differences exist between the source and target domains. We introduce the ADD module in UDA for RS images. The objective is to harness the distributional insights from source domains for efficient alignment, thereby minimizing the observed domain disparities. This strategy preserves significant semantic content from the source data, thereby enabling more effective training of the model on unlabeled target-domain data. Motivated by the insights from the latent diffusion model [21], a linear autoencoder is used [22] to compress the source-domain features, acting as the teacher, thereby enhancing computational efficiency. The diffusion model operates by adding noise into the potential vectors of the source-domain features (as the teacher). During the reversing phase, denoising is performed using the target-domain features (as the student), dynamically adjusting to match their noise characteristics for accurate predictions. The objective is to minimize the distance between the denoised target-domain features (student) and the latent representation of the source-domain features (teacher), achieved by employing a KD loss. This loss function captures the essence of the source-domain knowledge, enabling effective knowledge alignment.

In addition, existing UDA models predominantly rely on data-driven approaches, which often limit their adaptability to the inherent requirements of the methodology. Consequently, existing methodologies introduce a bias in the training process. Fig. 1 illustrates this phenomenon. The distribution of source domains includes many samples with much greenery in residential areas and only a few parking lots on roads. In contrast, the distribution of the target domain has more roadside residential samples and few green parking areas. As a result, the knowledge derived from the source domain tends to associate residential areas with greenery and roads with parking. During testing in the target domain, roadside residential samples are incorrectly categorized as parking, while green parking samples are mistakenly identified as residential areas. This phenomenon reflects the inherent bias of the model in domain knowledge, which forms the core challenge in addressing (UDA) tasks. To mitigate the influence of domain-specific knowledge on classification accuracy across various domains, the backdoor adjustment is utilized to address confounding factors. This allows us to train a classifier that maintains class consistency, thereby preventing the model from developing inherent biases. A concrete implementation of backdoor adjustment is carried out by proposing the CCI module. The binary cross-entropy loss is employed to simultaneously cluster similar sample pairs from both the source and target domains, effectively separating them by maximizing the distance between dissimilar pairs. This approach successfully avoids false correlations caused by environmental domain knowledge.

The main contributions of the article are given as follows.

1) A novel UDA framework based on diffusion distillation and causal inference is proposed.
2) To address the challenge of inconsistent data domain distributions, the ADD module is proposed. This module effectively narrows the domain gap between the source and target domains.
3) To eliminate the spurious correlation between environmental domain knowledge and causal features of samples, the CCI module is devised, which mitigates domain confusion bias.

II. Related Work

A. Diffusion Models

The diffusion model is a powerful generative model that can produce high-quality outputs in various domains such as

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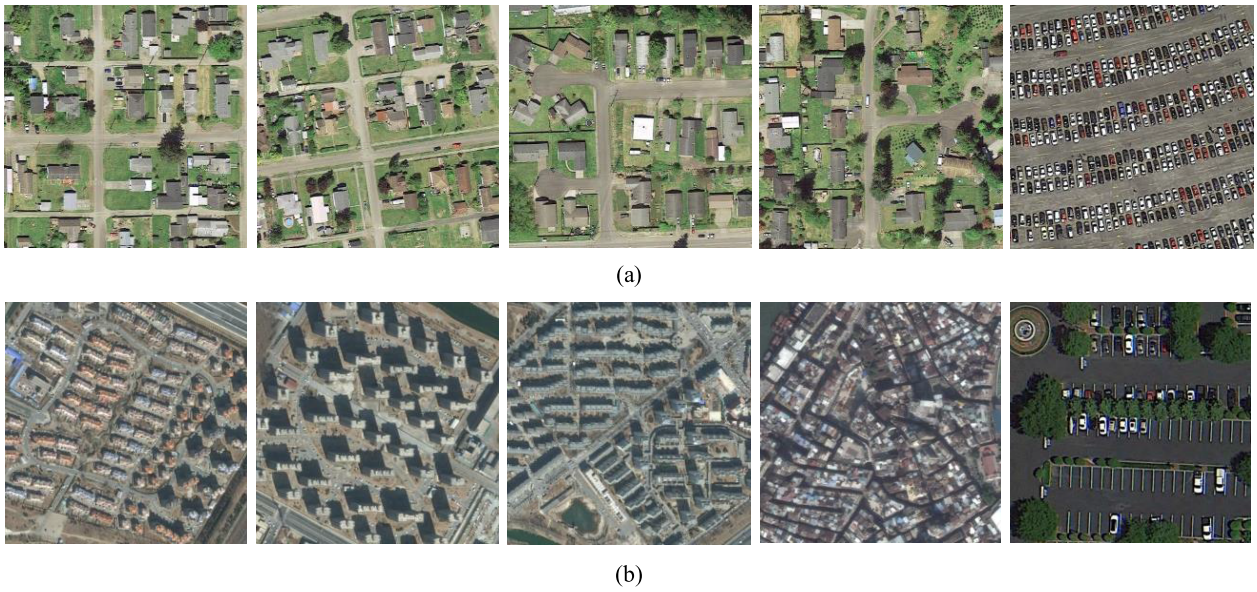


Fig. 1. Confounding effects of domain environment information on UDA. (a) Large number of green residential samples and a smaller number of roadside parking samples. (b) Large number of roadside residential samples and a smaller number of green parking samples.

images, text, and audio. It is a generative model that employs both forward and reverse processes to generate samples. In the forward process, noise is gradually added to the data, while, in the reverse process, the model attempts to predict the noise incrementally added at each step, gradually recovering the original, noise-free sample. Diffusion models are trained using the backpropagation algorithm in deep learning, but, fundamentally, they are Markov models. A key advantage of diffusion models is their ability to generate high-quality samples, backed by a solid theoretical foundation. They have also gained favor for their ability to generate highly realistic and diverse samples, breaking the long-standing dominance of generative adversarial networks (GANs) [23] in challenging image synthesis tasks. They have demonstrated exceptional performance in areas such as computer vision, natural language processing, time series, and multimodal.

Currently, the mainstream diffusion models are denoising diffusion probabilistic models (DDPM) [24]. In the field of optimization of probabilistic denoising diffusion models, the performance and efficiency of model generation have been significantly improved by fine-tuning the noise parameters and improving the noise injection method. Nichol and Dhariwal [25] improved the log-likelihood performance of the model by introducing specific cosine noise during the forward noising process. Strategies that enhance Markov chains have significantly boosted the model's performance in terms of sample generation speed and quality. DDIM [26] utilized predicted normal samples in each denoising step, which accelerated sampling speed significantly while maintaining sample quality by employing a non-Markov process. Numerous researchers are also exploring the combination of diffusion models with various learning strategies and network architectures to enhance the generation quality of the model and training efficiency. Sinha et al. [27] introduced contrastive self-supervised learning into the diffusion process, drawing on the idea of representation learning. Peebles and Xie [28] integrated diffusion models with the latest Transformer architecture. Many researchers are gradually beginning to use diffusion models in UDA tasks. Li et al. [29] proposed an unsupervised domain adapter that accomplishes the image-level domain alignment relying on the combination of the cross-domain style feature mixing and diffusion model. Peng et al. [30] proposed a novel domain-adaptive diffusion (DAD) module accompanied by a mutual learning strategy (MLS), which can gradually convert data distribution from the source domain to the target domain while enabling the classification model to learn along the domain transition process. Huang et al. [31] proposed a novel regularization method for domain-adaptive object detection, BlenDA, by generating the pseudosamples of the intermediate domains and their corresponding soft domain labels for adaptation training. The intermediate samples were generated by dynamically blending the source images with their corresponding translated images using an off-the-shelf pretrained text-to-image diffusion model. Xie et al. [32] proposed a self-awareness knowledge diffusion (SaKnD) method that leveraged self-knowledge without establishing cross-domain knowledge relationships. Benigmim et al. [33] leverage text-to-image diffusion models (e.g., stable diffusion) to generate a synthetic target dataset with photorealistic images.

We use diffusion models for distillation to help align features between two domains and reduce feature differences. The diffusion model is trained with source-domain knowledge to denoise the target-domain knowledge and align the features between the two domains. In addition, the KD is employed to transfer the source-domain knowledge to the target domain, thereby reducing the domain gap.

B. Causal Inference

Existing deep learning models rely on learning to generate predictions from massive data, which can inadvertently capture

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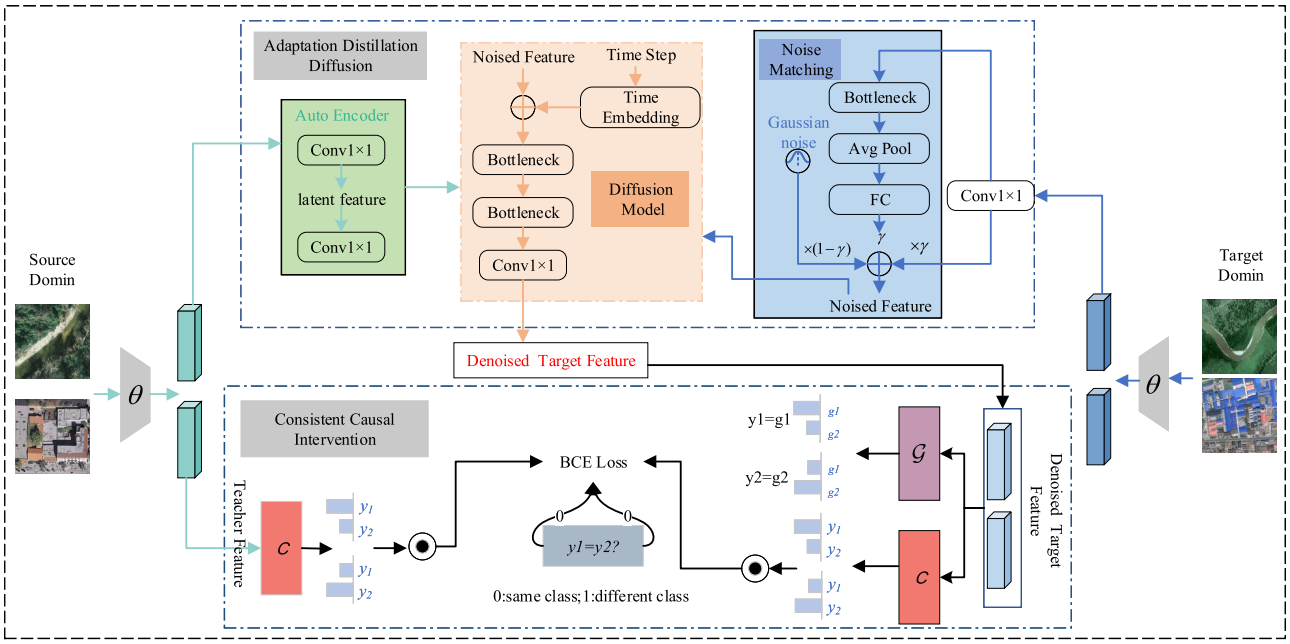


Fig. 2. Overall DDCI framework. Given paired source-domain samples, for each sample, features are extracted using the backbone θ to obtain F_{sou}. The features F_{sou} are processed using a linear autoencoder to obtain latent features Z_{sou}. The latent features undergo iterative noise addition and denoising to train the diffusion model. Similarly, for given target-domain samples, features are extracted using the backbone θ to obtain F_{tar}. The trained diffusion model is used to denoise F_{tar} with noise matching for adaptive noise matching, resulting in denoised target-domain features $\hat{F}_{\text{tar}}$. The paired denoised target-domain features and original source-domain features are input into the CCI module to eliminate confounding factor interference. We calculate the binary cross-entropy (BCE) loss of paired source- and target-domain samples to learn a classifier that is consistent with both source-domain classification and target-domain clustering. Since the target domain is unlabeled, clustering algorithms are used to generate pseudolabels for target-domain samples.

spurious correlations, leading to models that lack explainability, robustness, and generalization. Replacing correlation-based models with stable and interpretable causal models can mitigate the misleading effects of spurious correlations and overcome computational limitations. Causal inference, due to its low dependence on data and strong generalization, is increasingly incorporated into model algorithms. Recently, causal inference has seen growing application in various deep learning domains, such as visual representations [34], video processing [35], [36], [37], [38], visual language analysis [39], [40], explainability in deep learning [41], [42], and natural language processing [43]. These applications make it possible to construct models through deep causality, thus facilitating the process of reasoning and cognitive tasks.

To enhance the stability and invariance of machine learning models and improve their generalization capabilities, numerous approaches have emerged, which integrate causal learning with deep learning. The ability of causal learning is leveraged to establish strong causal relationships in complex and unstructured data, thereby boosting the explainability, generalization, and robustness of deep learning algorithms. In order to eliminate the effect of confounding factors in the data, Zhu et al. [44] proposed a causal-based approach that reweighted and resampled online data to enhance model performance. Feder et al. [45] delved into the significance of causal inference in natural language processing, highlighting its potential to boost model performance, enhance robustness, and facilitate explainability. Liu et al. [46] proposed a graph-based causal inference framework (CGI), which constructed causal graphs based on factual information to capture causal relationships between different variables. Wang et al. [47] introduced a causal attention module, which self-annotated confounders in an unsupervised manner and used causal intervention to remove the effects of confounders. The diffusion model is also gradually incorporated into causal learning. Sanchez and Tsaftaris [48] proposed a novel deep structured causal model on energy-based generative models [49]. Yue et al. [50] explored the potential of DDPMs for unsupervised learning of modular attributes. Causal learning is also being used in domain adaptation methods to improve the generalization of models. Zhu et al. [51] proposed a causal prototype-inspired contrast adaptation method to explore the invariant causal mechanisms between different high-resolution RS imagery domains and their semantic labels. Dai et al. [52] proposed a deep domain adaptation and disentangled representation network for unsupervised heterogeneous change detection, which bridges the domain gap from the perspective of causal mechanisms and compares the differences in the content feature space. Considering that semantic information is useful for the final task and domain information always indicates the discrepancy between two domains, Wang et al. [53] proposed to decouple the representations of semantic features from domain features to reduce domain bias.

We combine diffusion distillation with causal learning to improve domain adaptation. Diffusion distillation aligns features between the source and target domains to reduce the domain gap. However, it only focuses on the distribution of the source domain and ignores the inherent characteristics of the target domain. To address this, we use causal intervention to eliminate bias from domain-specific knowledge and learn

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classifiers that are consistent across both domains, enhancing the generalization performance of the model.

III. Methodology

To effectively tackle UDA, a novel framework is proposed, as illustrated in Fig. 2, which combines the ADD module and the CCI module. The overall framework will be introduced, and more importantly, the specific details of the two modules included in the framework will be discussed.

A. Overview

Given labeled source-domain RS image data $\mathcal{D}_{\text{sd}}$ with distribution p_{sd} and unlabeled target-domain RS image data $\mathcal{D}_{\text{td}}$ with distribution in p_{td}, the goal of UDA is to find a classifier $\mathcal{C}$ using the labeled source-domain data $\mathcal{D}_{\text{sd}}$ such that it can effectively transfer to the unlabeled target domain $\mathcal{D}_{\text{td}}$ and achieve good performance. In this article, a new framework DDCI is proposed for UDA, as shown in Fig. 2.

On the one hand, the distributional transformation capability of the diffusion model is leveraged to bridge the domain gap between the source and target domains. Specifically, backbone is first used to obtain the features f_{sd} and f_{td} in the source and target domains on $\mathcal{D}_{\text{sd}}$, respectively. Due to the high dimensionality of the features, a linear autoencoder is employed [54] to generate latent space vectors z_{sd}. Based on the latent diffusion probability model, noise addition and denoising are performed in the source-domain latent space vector z_{sd}. Then, the target-domain features are fed into the denoising inference process to obtain the target-domain features $\hat{f}_{\text{td}}$ after denoising. It is worth noting that in UDA, the aim is to minimize the domain gap between the two domains, by leveraging source-domain knowledge to aid target model training and reducing generalization error in the target domain. Therefore, inspired by [17], the source-domain features are considered the teacher features and the target-domain features the student features, attributing the domain gap to noise in the target domain. After the denoising inference using the target-domain feature f_{td} denoising produces the denoising feature $\hat{f}_{\text{td}}$, the KL distance between the denoising feature $\hat{f}_{\text{td}}$ and the latent vector z_{sd} is computed (i.e., the source-domain feature and the denoised target-domain features for knowledge distillation) to train the constraints on the diffusion model learning noise process. This effectively transfers valuable source-domain knowledge for target model training, significantly aligning source and target features and, thereby, reducing the domain gap.

On the other hand, the structural causal model (SCM) is constructed for UDA. As shown in Fig. 3(a), D denotes the domain, S denotes the domain semantic features, X denotes the RS images, and Y represents their corresponding labels. $X \rightarrow Y$ signifies the RS image classification task, which is influenced by unobserved domain semantic features S. $S \rightarrow X$ signifies the generation of pixel-level image samples, and $S \rightarrow Y$ represents the process of defining semantic classes. It is the marginal distribution $P(S|D = \text{sd}) \neq P(S|D = \text{td})$ that leads to the conditional distribution $P(Y|X, D = s) \neq P(Y|X, D = \text{td})$, which causes the domain gap. To minimize the domain gap, it is necessary to eliminate the spurious correlation between the environmental domain knowledge encoded in S and the causal features of the sample. According to the causal structure diagram established in Fig. 3(a), to eliminate the influence of domain knowledge S on the classification result Y, it is necessary to block the path between S and X. In SCM, intervening in a variable means removing all arrows pointing to that variable, thereby breaking the natural causal relationship of that variable. Therefore, intervening in X can eliminate the arrow pointing to X, blocking the pathway between S and X, as shown in Fig. 3(b). The backdoor adjustment method is used in causal inference to eliminate the effect of confounding factor S on X, denoted as

$$P(Y = y|\text{do}(X = x), D) = \sum_{d \in \text{sd,td}} P(Y = y|X = x, D = d)P(D = d) \quad (1)$$

where D represents different domains, sd represents the source domain, td represents the target domain, and $\text{do}(X = x)$ represents intervening in x to eliminate the confusion caused by domain environment knowledge.

The CCI module is proposed to implement backdoor adjustment. Specifically, in the source domain, BCE loss is computed for paired samples using source-domain features f_{sd} and denoised target-domain features $\hat{f}_{\text{td}}$. In the target domain, clustering is first performed, followed by BCE loss calculation using cluster labels for paired samples. Ultimately, a class-invariant consistent classifier is learned by constraining the consistency of the source- and target-domain classifications. This approach respects the distinctiveness of source-domain data while considering the inherent characteristics of target-domain data. As a result, our model becomes less reliant on raw data and, instead, focuses on learning the fundamental causal principles underlying samples. This, in turn, prevents confusion that might arise from domain-specific knowledge.

Through the DD and CCI modules, the UDA problem is successfully addressed. In the following, a detailed description of these modules will be provided.

B. Adaptation Diffusion Distillation

Diffusion models can convert Gaussian distributions into arbitrary specific distributions. The diffuse distributed boundaries capability of the diffusion model can be utilized to transform between domains and reduce domain gaps. However, for domain adaptation methods, it is crucial to preserve semantic information from the source domain to assist the target-domain model. Thus, the ADD module is proposed. This module combines diffusion models with distillation, not only utilizing the transformation ability of diffusion models to reduce domain gaps but also facilitating the transfer of knowledge from source- to target-domain data, enabling the model to extract shared features across domains, ultimately enhancing its performance on target data. First, features from the source and target RS images are extracted using a backbone. Then, the diffusion model is employed to perform explicit denoising and matching on these features. Ultimately, the distilled valuable

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information from the teacher representation guides the student model to produce better student representations. Specifically, given the source-domain RS data feature (i.e., teacher feature) f_{sd} obtained from the backbone and the target-domain RS data feature f_{td} (i.e., student feature), during the forward diffusion process, noise is added to the teacher features f_{sd}, denoted as

$$q\left(f_k^{\mathrm{sd}} \middle| f_{k-1}^{\mathrm{sd}}\right) = \mathcal{N}\left(f_k^{\mathrm{sd}}; \sqrt{\bar{\alpha}_k} f_{k-1}^{\mathrm{sd}}, \left(1-\bar{\alpha}_k\right)\mathbf{I}\right) \tag{2}$$

where $k \in \{1, 2, \ldots, K\}$. f_{k-1}^{sd} is the feature map already diffused for $(k-1)$ steps, which is the input of q, and f_k^{sd} is the output of the diffusion operator q. The notation for directly sampling f_k^{sd} at arbitrary time step with noise variance schedule β is denoted as

$$\alpha_k := 1 - \beta_k, \quad \bar{\alpha}_k := \prod_{j=1}^{k} \alpha_j. \tag{3}$$

Therefore, we can express f_k^{sd} as a linear combination of f_0^{sd} and noise variable ϵ_k, denoted as

$$f_k^{\mathrm{sd}} = \sqrt{\bar{\alpha}_k} f_0^{\mathrm{sd}} + \sqrt{1-\bar{\alpha}_k}\epsilon_k. \tag{4}$$

A diffusion model with a loss function L_{diff} is trained, denoted as

$$\mathcal{L}_{\mathrm{diff}} = \left\|\epsilon_k - \mu_\theta\left(f_k^{\mathrm{sd}}, k\right)\right\|_2^2 \tag{5}$$

where f_k^{sd} is the input feature map of the reverse operator p_θ and f_{k-1}^{sd} is the output. $\mu_\theta(f_k^{\mathrm{sd}}, k)$ is the mean value of the output feature distribution, which can be predicted by the reverse operator p_θ, denoted as

$$\mu_\theta\left(f_k^{\mathrm{sd}}, k\right) = \frac{1}{\sqrt{\alpha_k}}\left(f_k^{\mathrm{sd}} - \frac{\beta_k}{\sqrt{1-\bar{\alpha}_k}}(f)_\theta\left(f_k^{\mathrm{sd}}, k\right)\right) \tag{6}$$

where f_θ is the learnable neural network within the reverse operator p_θ.

The student features are then fed into the learned diffusion model for iterative denoising to learn the diffusion model, denoted as (7). The target-domain feature f_{td} is the initial noisy feature of the process

$$\hat{f}_{\mathrm{td}} = p_\theta\left(f_{k-1}^{\mathrm{td}} \middle| f_k^{\mathrm{td}}\right) = \mathcal{N}\left(f_{k-1}^{\mathrm{td}}; \mu_\theta\left(f_k^{\mathrm{td}}, k\right), \beta_k\mathbf{I}\right). \tag{7}$$

After the denoising process, the denoised target features $\hat{f}_{\mathrm{td}}$ are obtained, and the KD loss is calculated between the denoised student features and the original teacher features, denoted as (8). The target-domain features denoised by the diffusion model are very similar to the source-domain features, which ensures that our distillation is performed more consistently

$$\begin{aligned}\mathcal{L}_{\mathrm{kd}} &= t^2 \times \mathrm{KL}\left(f_{\mathrm{sd}}, \hat{f}_{\mathrm{td}}\right) \\ &= t^2 \sum f_{\mathrm{sd}} \log \frac{f_{\mathrm{sd}}}{p_\theta\left(f_{k-1}^{\mathrm{td}} \middle| f_k^{\mathrm{td}}\right)}\end{aligned} \tag{8}$$

where t represents the temperature parameter of distillation.

However, the training process of the diffusion model necessitates μ_θ to conduct K forward passes for denoising target-domain features, along with one pass for training using source-domain features. As a result, denoising with the diffusion model involves a higher computational cost. To reduce the computational cost of the diffusion model, a lightweight diffusion model is used. The idea of the latent diffusion model is followed, which involves adding noise and denoising in the latent vector space. For source-domain features, a linear autoencoder is employed to compress the feature channels, generating a latent vector z_{sd} through the encoder. The encoder-produced z_{sd} is then used to reconstruct the source-domain features via the decoder. The latent vector z_{sd} is employed to train the diffusion model. When training the diffusion model using the latent vector z_{sd}, backpropagation is performed without computing the gradient. To calculate the distillation loss between target-domain features and source-domain features, the target-domain features are linearly projected to have the same dimension as the latent vector. These projected target-domain features are then fed into the diffusion model for a reverse denoising process (8), ultimately producing the denoised target-domain features (student features). In order to distill the knowledge of the source-domain features, the denoising process of the target-domain features is supervised by a distillation loss between the denoised target-domain features and the latent features.

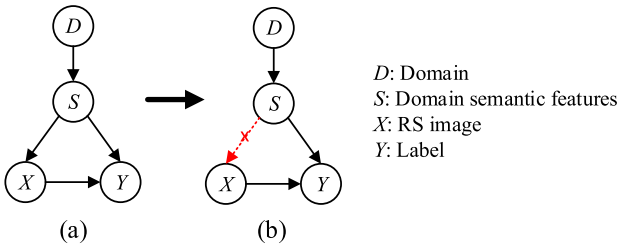


Fig. 3. (a) SCM for UDA of RS. (b) Intervene in X, block the backdoor path, and eliminate the influence of domain prior knowledge on the results.

However, our inability to accurately estimate the noise level gap between the source-domain features and the target-domain features makes it impossible to determine at which initial time step, the diffusion process should begin. To solve this problem, the noise adaptation module is proposed to adapt the noise level of the target-domain features to a predefined noise level. As shown in Fig. 1, a hyperparametric learning weight γ is designed to fuse the Gaussian noise and the target-domain features such that the target-domain features and the noise features are adaptively matched at the initial time step of the reverse denoising process. Thus, the initial noise signature in the denoising process of the diffusion model becomes

$$z_K^{(\mathrm{td})} = \gamma z^{(\mathrm{td})} + (1-\gamma)\epsilon_K. \tag{9}$$

In the diffusion model denoising process, the denoised target-domain features are produced only when the target-domain features and the noise level are matched less different from the source-domain features, so the source-domain knowledge is well transferred to the student model to help produce better target-domain features.

C. Consistent Causal Intervention

As shown in Fig. 3(a), the classification results are affected by domain prior knowledge due to a backdoor path $X \leftarrow S \leftarrow D \rightarrow S \rightarrow Y$ between image X and label Y. To eliminate

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TABLE I

UDA RESULT ON 12 TASKS ACROSS FOUR DATASETS. BOLD NUMBERS INDICATE THE BEST RESULTS, WHILE UNDERLINED NUMBERS INDICATE THE SECOND RESULT. "HYBRID" REPRESENTS A HYBRID METHOD BASED ON ADVERSARIAL, CONTRASTIVE, AND DISTILLATION TECHNIQUES

Method	Based Method	U2W	U2A	U2R	W2U	W2A	W2R	A2U	A2W	A2R	R2U	R2W	R2A	Avg
ResNet [55]	Aligned	71 .2	66.4	56.5	75.1	93.1	71.3	74.7	98.4	76.9	80.7	86.7	85.2	78.0
DDC [5]	Aligned	71.8	66.3	57.7	78.0	90.1	70.1	78.3	99.2	75.6	79.7	89.2	84.7	78.4
DCORAL [6]	Aligned	73.7	66.0	56.5	79.4	92.6	72.8	79.3	99.4	77.0	83.4	89.9	85.5	79.6
MRAN [7]	Aligned	78.3	83.0	64.8	81.7	94.1	79.2	84.7	98.7	79.0	83.1	93.4	93.9	84.5
AMRAN [8]	Aligned	89.6	84.8	69.0	85.2	94.8	79.5	88.2	**99.7**	81.0	87.6	94.6	94.5	87.4
WPS-DSA [9]	Aligned	94.0	90.3	71.0	88.1	95.5	80.7	81.3	**99.7**	83.8	88.6	95.9	96.5	88.8
RevCGrad [10]	Adversarial	72.5	67.5	55.8	80.9	92.3	72.8	80.7	99.1	76.8	83.9	90.2	85.5	79.8
JAN [11]	Adversarial	76.6	69.2	61.9	81.9	92.3	73.6	82.9	98.1	78.1	81.6	88.6	84.3	80.8
DASNET [12]	Adversarial	92.1	76.3	60.6	83.1	95.8	76.0	82.6	99.4	77.0	88.7	95.9	90.5	84.8
TADA [13]	Adversarial	93.1	90.6	67.5	89.8	96.6	**87.1**	88.4	99.3	84.2	88.9	96.7	97.1	89.9
MRDAN [14]	Adversarial	96.1	90.8	**78.6**	89.1	96.4	87.0	89.1	99.8	85.1	90.9	97.4	96.9	91.4
DAN [15]	Contrastive	74.7	67.1	57.5	80.1	92.8	72.8	80.3	99.4	76.9	82.9	88.9	85.3	79.9
SRKT [16]	Hybrid	94.0	95.3	74.5	88.3	97.0	85.6	84.8	98.7	**85.3**	89.2	96.8	**98.4**	90.1
DDCI(ours)	Hybrid	**96.5**	**98.6**	75.3	**96.5**	**98.3**	79.2	**89.3**	99.0	82.0	**91.3**	**98.4**	97.7	**91.5**

the spurious correlation between the environmental domain knowledge and the causal features of sample identity markers caused by the confounding factor, the backdoor adjustment is employed, denoted as (10). We intervene in X to eliminate the influence of domain prior knowledge on X, thereby blocking the backdoor path. This can make the classification result depend only on the essential features of X, as shown in Fig. 3(b)

$$\begin{aligned} &P(Y=y|\mathrm{do}(X=x),D) \\ &=\sum_{d\in\mathrm{sd,td}} P(Y=y|X=x,D=d)P(D=d) \\ &=P(Y=y|X=x,D=\mathrm{sd})P(D=\mathrm{sd}) \\ &\quad+P(Y=y|X=x,D=\mathrm{td})P(D=\mathrm{td}). \end{aligned} \tag{10}$$

To not only respect the specificity of source-domain data but also account for the inherent distribution of target-domain data, the aim is to learn a classifier that aligns with both source-domain classification and target-domain clustering. The BCE loss is introduced, which encourages the model to cluster samples with the same label more closely and pushes apart those with different labels based on their similarity. By jointly minimizing source- and target-domain losses, the model learns a domain-agnostic classifier that makes predictions solely based on causal features. Specifically, the BCE loss is employed, denoted as $\mathrm{BCE}(\mathrm{sd},\mathcal{C})$, to ensure consistency in source-domain data classification, denoted as

$$\mathrm{BCE}(\mathrm{sd},\mathcal{C})=\mathbb{E}_{(\mathbf{x}_i,\mathbf{x}_j)\sim\mathrm{sd}}\left[b\log\left(\hat{y}\right)+(1-b)\log\left(1-\hat{y}\right)\right] \tag{11}$$

where $b:=\mathbb{1}(y_i=y_j)$ is an indicator function, taking a value of 1 when $(\mathbf{x}_i,\mathbf{x}_j)$ belongs to the same class and 0 otherwise. The predicted similarity is denoted as $\hat{y}=\mathcal{C}(\mathbf{x}_i)^{\mathrm{T}}\mathcal{C}(\mathbf{x}_j)$.

Since the target-domain data lack labels, it is not possible to compute $\mathrm{BCE}(\mathrm{td},\mathcal{C})$ directly as is done for the source domain. Therefore, the target-domain data are clustered, and the cluster labels are used to estimate $\mathrm{BCE}(\mathrm{td},f)$, denoted as

$$\mathrm{BCE}(\mathrm{td},\mathcal{C})=\mathbb{E}_{(\mathbf{x}_i,\mathbf{x}_j)\sim\mathrm{td}}\left[b'\log\left(\hat{y}\right)+\left(1-b'\right)\log\left(1-\hat{y}\right)\right] \tag{12}$$

where $b':=\mathbb{1}(\mathrm{argmax}\ \mathcal{G}(x_i)=\mathrm{argmax}\ \mathcal{G}(x_j))$ and $\mathcal{G}$ is the clustering head.

Consistency in target-domain classification is achieved by comparing cluster labels. In addition, to reduce computational complexity, minibatch sampling is adopted for online clustering using the rank-statistics algorithm. Since clustering is solely performed on the target domain, it effectively captures the target domain's data distribution without being influenced by the source domain's distribution.

The consistency of the CCI module is manifested in its ability to enforce consistent decision boundaries for classifying across both the source and target domains. The objective is denoted as

$$\mathcal{L}_{\mathrm{CCI}}=\min_{\theta,\mathcal{C}}\mathrm{BCE}(\mathrm{sd},\mathcal{C})+\mathrm{BCE}(\mathrm{td},\mathcal{C}). \tag{13}$$

The BCE loss training of the classifier aims to make the pairwise label similarities consistent between the source and target domains. This constraint demands that the backbone network θ generates an invariant classifier c, which minimizes BCE losses simultaneously in both domains. This constraint prevents a situation where one loss is minimized at the expense of the other, striving for a lower combined sum of both objectives.

D. Loss Function

During the training of the autoencoder, the loss is the mean square error (MSE) between the original source-domain features and the reconstructed source-domain features, denoted as

$$\mathcal{L}_{\mathrm{ae}}=\left\|\hat{f}^{(\mathrm{sd})}-f^{(\mathrm{sd})}\right\|_2^2. \tag{14}$$

The latent representation of the teacher $z^{(\mathrm{sd})}$ and the denoised student representation $\hat{f}^{(\mathrm{td})}$ are then used to compute the KD loss and supervise the students, as illustrated in (8). The source-to-target-domain classification consistency loss is denoted as (13). Diffusion model loss during training is denoted as (5). Therefore, the overall loss function is

$$\mathcal{L}_{\mathrm{overall}}=\lambda_1\mathcal{L}_{\mathrm{ae}}+\lambda_2\mathcal{L}_{\mathrm{kd}}+\lambda_3\mathcal{L}_{\mathrm{CCI}}+\lambda_4\mathcal{L}_{\mathrm{diff}} \tag{15}$$

where λ_i represents the weighting parameter for each loss.

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TABLE II
UDA ACC RESULT ON 12 TASKS ACROSS FOUR DATASETS

ACC	U2W	U2A	U2R	W2U	W2A	W2R	A2U	A2W	A2R	R2U	R2W	R2A
farmland	94.0	98.1	94.3	99.0	99.2	96.8	99.0	98.0	96.8	99.0	98.0	98.4
forest	69.8	100.0	97.3	100.0	100.0	98.0	98.0	98.1	85.3	99.0	94.3	100.0
industrial	100.0	97.2	89.0	100.0	97.2	38.8	77.0	98.1	51.3	86.0	100.0	97.4
parking	98.0	99.4	38.5	99.0	99.7	92.5	98.0	100.0	73.0	100.0	98.0	99.0
residential	94.4	98.0	40.8	83.0	97.8	56.5	64.0	100.0	82.0	66.0	100.0	93.7
river	100.0	99.0	91.8	98.0	96.1	92.8	100.0	100.0	97.8	98.0	100.0	98.0
mean	92.7	98.6	75.3	96.5	98.3	79.2	89.3	99.0	81.0	91.3	98.4	97.7

TABLE III
UDA IOU RESULT ON 12 TASKS ACROSS FOUR DATASETS

IOU	U2W	U2A	U2R	W2U	W2A	W2R	A2U	A2W	A2R	R2U	R2W	R2A
farmland	71.2	97.3	90.4	99.0	96.1	93.0	99.0	98.0	84.7	98.0	98.0	96.3
forest	69.8	94.3	94.4	98.0	99.2	95.1	97.0	96.3	84.4	97.1	94.3	99.6
industrial	93.0	99.2	52.6	74.1	95.0	27.5	44.8	98.1	41.0	51.8	100.0	89.6
parking	98.0	95.7	24.1	99.0	97.5	54.0	98.0	100.0	48.5	98.0	98.0	98.2
residential	91.1	97.6	40.0	82.6	96.6	56.2	57.1	98.1	78.5	61.7	98.2	98.5
river	98.2	97.4	83.6	98.0	95.4	88.3	97.0	98.2	82.3	97.0	93.3	97.3
mean	86.9	96.9	64.2	91.8	96.6	69.0	82.2	98.1	69.9	83.9	97.0	95.6

IV. EXPERIMENT

In this section, the proposed novel framework DDCI is evaluated. Domain adaptation scene classification tasks are constructed using four existing datasets: the UC Merced Land Use Dataset (UCM) [56], the Wuhan University (WHU)-RS Dataset [57], the Aerial Image Dataset (AID) [58], and the RSSCN7 Dataset [59]. All tasks are configured in a source-to-target-domain adaptation setting.

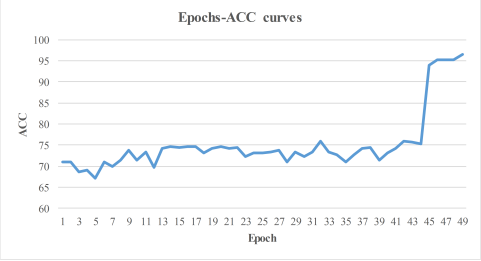


Fig. 4. Epoch–ACC curves during the testing process.

A. Dataset Overview

To better facilitate the experiment and present the findings, the benchmark dataset provided by [8] is employed. Four RS image datasets were utilized, where each pair was composed of two datasets, serving as source and target domains, respectively, for the task of UDA.

1) UCM: This dataset is a classic RS scene recognition dataset, specifically for classifying land use scenarios in urban areas. The images were extracted from the USGS National Map. Characterized by a spatial resolution of around 0.3 m, each image measures 256 × 256 pixels. The dataset contains 21 categories of scenes with 100 images in each category, totaling 2100 images. The dataset exhibits a limited diversity in scene types, presenting a challenge to domain adaptation methods due to the presence of similar objects and texture patterns.

2) WHU: This dataset is a set of satellite images exported from Google Earth, which provides high-resolution satellite images up to 0.5 m. It is suitable for scene classification and retrieval. The image scale is 600 × 600 pixels and contains 19 classes of scenes, 50 per class, totaling 1005 images. Images within the same class were collected from different regions of satellite images with varying scales, orientations, and lighting conditions, presenting a challenge for domain adaptation methods due to their inherent diversity.

3) AID: This dataset is a large aerial image collection, consisting of sample images sourced from Google Earth, with a resolution of 600 × 600 pixels. It contains 30 classes of scenes, with about 200 images per class, for a total of 10 000 images of labeled scenes. The diversity arises due to the global origin of the sample images and varying imaging conditions, resulting in significant interclass variation and increased intraclass diversity.

4) RSSCN7: The dataset consists of 2800 RS images, distributed across seven representative scene categories. Each image in the dataset has a pixel size of 400 × 400. The images are sourced from various seasons and weather conditions and sampled at different scales, which contributes to its inherent diversity and presents a significant challenge for domain adaptation tasks.

Following the approach in [8], the images from six common classes across the four challenging datasets are selected, including residential, farmland, forests, industrial, parking, and rivers.

B. Experiment Settings

The four datasets (UCM as U, WHU as W, AID as A, and RSSCN7 as R) are paired in twos, designating one as the

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TABLE IV
RESULTS OF ABLATION EXPERIMENTS

Backbone	CCI	ADD	U2W	U2A	U2R	W2U	W2A	W2R	A2U	A2W	A2R	R2U	R2W	R2A	Avg
✓			71.2	66.4	56.5	75.1	93.1	71.3	74.9	98.4	76.9	80.7	86.7	85.2	78.0
✓	✓		87.5	94.9	73.3	86.3	93.4	68.6	70.3	99.0	81.0	91.2	97.1	97.1	86.6
✓	✓	✓	**92.7**	**98.6**	**75.3**	**96.5**	**98.3**	**79.2**	**89.3**	**99.0**	**82.0**	**91.3**	**98.4**	**97.7**	**91.5**

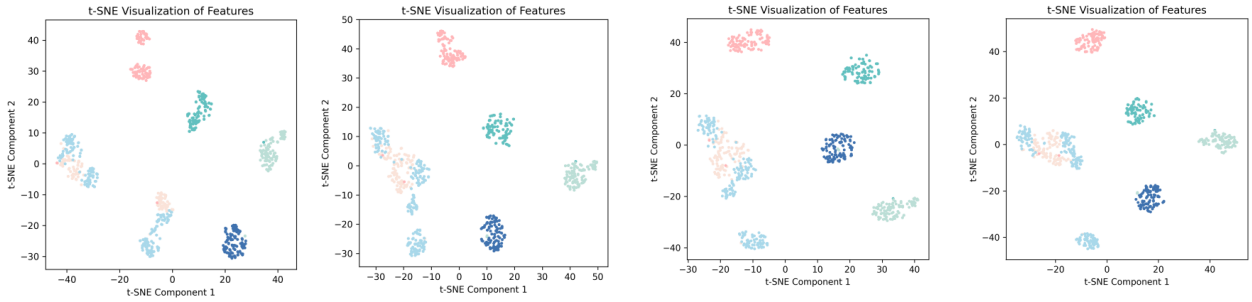


Fig. 5. Visualization of the features corresponding to the ACC curves.

source domain and the other as the target domain, resulting in a total of 12 possible combinations: U2W, U2A, U2R, W2U, W2A, W2R, A2U, A2W, A2R, R2U, R2W, and R2A.

The PyTorch framework is employed for training, utilizing a ResNet50 backbone. To enhance cross-domain features, the model is fine-tuned with diffusion distillation. The classifier is trained using causal intervention with backpropagation to ensure consistent classification results between the source and target domains. The batch size is set to 28, training is conducted for 50 epochs, and a clustering feature dimensionality reduction parameter of 50 is used, with a diffusion step size K set to 1000. The KD is performed on source- and target-domain features, utilizing an autoencoder loss $\mathcal{L}_{\text{ae}}$ and a KD loss $\mathcal{L}_{\text{kd}}$, both with weight parameters set to 0.1. The classifier is trained with the CCI loss $\mathcal{L}_{\text{CCI}}$, with a weight parameter of 1.0. All experiments are conducted on a single NVIDIA RTX A6000 GPU.

C. Evaluation and Results

Our method was implemented on four datasets, and the comparative results with other approaches are presented as follows.

1) Evaluation Metrics: Our evaluation criterion is the accuracy with which models trained on the source domain classify samples from the target domain. The average precision (ACC) and the intersection over union (IOU) are computed, which is the ratio of the intersection between the predicted and true values for a particular class to their union.

2) Results: Our results on 12 RS image domain adaptation tasks are shown in Table I, where the bolded ones are the best results. Our method significantly outperforms other UDA approaches in terms of accuracy. In the following, a detailed analysis of the results is presented.

Among the 12 tasks of domain adaptation for all cross-domain RS images compared to [8], our method improved by 7.75%, 16.27%, 9.10%, 13.26%, 3.67%, 1.29%, 1.23%, 4.27%, 4.03%, and 3.41% on ten tasks, respectively, while only decreasing by 0.38% and 0.68% (barely a decline) in two tasks. The overall accuracy across all tasks increased by 4.77%, indicating that the diffusion model's alignment of source- and target-domain features effectively reduced domain gaps, and CCI led to more accurate classifications. The detailed classification results for each task's six categories are illustrated in Tables II and III.

As shown in Table I, our approach outperforms significantly in most tasks, achieving the highest performance, particularly in tasks such as U2W, U2A, U2R, W2U, W2A, and R2W. This indicates that our method is more effective in handling data transformation between different domains, and it holds a clear advantage when tackling UDA problems. Although our method is not the top performer in W2R and A2R tasks, it still exhibits commendable performance, with only a margin of 0.38% and 0.68% behind the best. In general, our method effectively bridges the domain gap and reduces reliance on domain priors. The ADD module aligns features between the source and target domains through a denoising process, avoiding the instability of adversarial training. The CCI module eliminates confusion from domain priors, enabling simultaneous adaptation to both domain distributions. This leads to more stable performance in UDA. By directly transferring knowledge without the complexity of adversarial or contrastive learning, our method simplifies training and improves efficiency, offering a clear advantage over methods that struggle with biased priors or feature misalignment.

Tables II and III present the classification metrics of different categories in various RS image domain adaptation scenarios, including ACC and IOU. In most scenarios, our method achieved high classification metrics for categories such as farmland, forest, industrial, parking, and residential. In more than half of the 12 domain adaptation tasks, both ACC and IOU exceeded 90% in domain adaptation scenarios. For different categories, our method exhibited different advantages in different RS image scenes. For example, in the farmland scene, our method achieved high accuracy in almost all categories,

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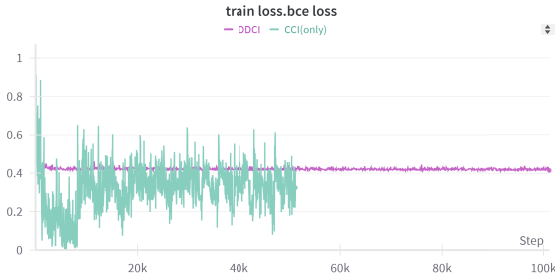


Fig. 6. BCE loss during training.

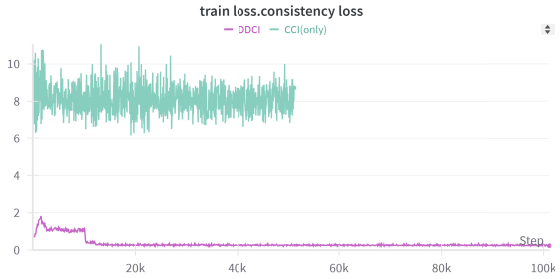


Fig. 7. Loss of consistency between source and target domains during training.

while, in the industrial scene, our method excelled in certain categories. On the average metric (mean), our method obtained high ACC and IOU metrics, further indicating the robustness and generalization ability of our method in different scenarios. Overall, our method achieved high evaluation metrics in multiple scenarios, demonstrating good classification capability. Meanwhile, our method exhibited good robustness in different RS image scenes, capable of adapting to different data distributions and categories. Our method performed well on average metrics, indicating excellent generalization ability and robustness, suitable for various task scenarios. Therefore, our method demonstrates promising performance and application prospects in solving domain adaptation problems.

3) Ablation Experiments: As shown in Table IV, to validate the effectiveness of our two primary modules, ablation experiments are conducted to analyze them. The ResNet50 [55] is used without any module as a baseline and observes significant improvements in classification results after incorporating causal intervention and diffusion measures into the backbone. The addition of the CCI module led to excellent performance in multiple RS domain adaptation tasks, effectively removing the influence of data-driven domain prior knowledge and enhancing model generalization. However, it can be observed that on the W2R and A2U tasks, where the domain gap is too large, the performance still needs to be improved. To address this, the diffusion model is introduced to enhance feature transformation across different data distributions. Adding the ADD module further improved classification accuracy by adaptively distilling and diffusing source-domain features for target-domain feature extraction. The overall model demonstrated strong performance across 12 task scenarios, showcasing good generalization and robustness in solving UDA of RS problems.

TABLE V
TIME COST OF DIFFERENT METHODS ON TASK U2A

Method	Training(s)	Test(s)	Accuracy(%)
ResNet50 [55]	4.125	9.530	78.0
DCORAL [6]	9.343	28.768	79.6
DDC [5]	9.353	28.508	78.4
DAN [15]	9.467	28.548	79.9
RevCGrad [10]	12.884	27.700	79.8
MRAN [7]	9.494	28.633	84.5
AMRAN [8]	9.627	28.890	87.4
DDCI(ours)	16.140	34.85	91.5

4) Visualization of Results: The effectiveness of the method is further demonstrated through visualizations. As shown in Fig. 4, with the increase in epochs, the accuracy of RS image scene classification is constantly improving. The classification errors between industrial and parking, as well as residential and parking, are gradually decreasing, effectively eliminating the confusion caused by domain priors. As the training progresses, the parameters of the model are continuously adjusted to minimize the loss function, leading to the learning of dataset patterns and invariance. This improves classification accuracy and reduces misclassification, making the result better and better. From the visualization of the feature map in Fig. 5, it can be observed that the feature classification improves as the number of trainings increases. The intraclass distances shrink, while interclass distances grow. The domain gap is well narrowed by the CCI module, which improves the accuracy of the classification.

In addition, the effectiveness of ADD is validated by comparing loss results with and without the ADD module. As shown in Figs. 6 and 7, the loss significantly decreases after incorporating the ADD module. In Fig. 6, the BCE loss becomes more stable. Fig. 7 shows a reduction in consistency loss.

5) Computational Complexity Analysis: The training time for an epoch is shown in Table V. The adaptation method that only uses ResNet50 for training on the source domain without considering the target domain is the least time-consuming. The proposed DDCI method inevitably increases the training and testing times due to diffusion models. However, too much time indicates that the model performs a significant amount of computation and can generate more detailed and informative feature maps, which are very useful for image recognition, natural language processing, or other complex tasks. It can be calculated that ResNet-50 has 25 557 032 parameters (about 25.56 M) and AMRAN has 26 819 760 parameters (about 26.82 M). The parameter size of the proposed DDCI method is 27.745M, which is 3.33% more than that of the baseline method. However, the effect increased by 4.48%. From the number of parameters, it can be seen that the model may have a high ability to learn complex patterns and representations from data. This is particularly beneficial for tasks involving subtle differences or large detailed datasets. Moreover, larger models with more parameters are sometimes more robust to overfitting, especially when combined with regularization techniques and sufficient training data.

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Therefore, our new framework effectively addresses the challenge of UDA for RS, which is of great significance and value.

V. Conclusion

This study introduces an innovative UDA framework designed to bridge the domain discrepancy in RS data. Central to this framework are two critical components: the adaptive diffusion distillation (ADD) module and the CCI module. The ADD module is proposed to reduce the domain gap between the source domains and target domains by employing diffusion distillation, which aligns the feature representations across different domains. This process enables the target-domain model to assimilate shared semantic insights from both domains, enhancing its learning capabilities. Meanwhile, the CCI module can effectively mitigate domain spurious correlations between environmental domain knowledge and causal features of RS scenes. It achieves this by training a consistent classifier that concurrently considers the classification of the source domain and the clustering of the target domain, thus fostering a more robust understanding of the data. Our framework has been successfully implemented in the context of RS scene classification, with extensive experiments across four well-established datasets confirming its efficacy. Looking ahead, we aim to investigate more efficient model architectures and strategies to further advance the field of UDA, ensuring that our methods remain at the cutting edge of RS image analysis.

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